HCov Exam Sheet 2025/26 - The Gratwick Files

Theorem: Go-To Theorems

- **Triangle Inequality** $|z+w| \le |z|+|w|$
- **Reverse Triangle Inequality** $|z-w| \ge ||z|-|w||$
- **M-L Lemma** For regular Γ, continuous $f:\Gamma\to\mathbb{C}$: $\left|\int_\Gamma f(z)\ \mathrm{d}z\right| \le \max_{z\in\Gamma}|f(z)|\ell(\Gamma)$

Complex Fundamentals

Definition: Basics

- $\mathbb{C}=\{x+iy|x,y\in\mathbb{R}\}$ is a field
- $\Re(z)=\frac{z+\overline{z}}{2}$, $\Im(z)=\frac{z-\overline{z}}{2i}$
- $z=re^{i\theta}$: $x=r\cos\theta$, $y=r\sin\theta$
- $e^{in\theta}=(e^{i\theta})^n$
- $|z|=0 \iff z=0$
- $\left|z^2\right|=|z|^2=z\overline{z}$
- $\overline{z\overline{w}}=(\overline{z})(\overline{w})$

Definition: The Argument

- For some $z=re^{i\theta}$, θ is an *argument* of z. $\mathrm{Arg}(z)$ is the unique $\theta\in(-\pi,\pi]$
- $\arg(z)=\{\theta:z=|z|e^{i\theta}\}=\{\mathrm{Arg}(z)+2k\pi:k\in\mathbb{Z}\}$
- $\arg(zw)=\arg(z)+\arg(w)$, $\arg(\overline{z})=-\arg(z)$
- $\mathrm{Arg}(zw)=\mathrm{Arg}(z)+\mathrm{Arg}(w)$, $\mathrm{Arg}(\overline{z})=-\mathrm{Arg}(z)$ (both potentially $\pm 2k\pi$ for some $k\in\mathbb{Z}$)
- $\mathrm{Arg}_\phi(z)$ is the branch of $\arg(z)$ s.t $\phi<\mathrm{Arg}_\phi(z)\le\phi+2\pi$

Theorem: Properties Of Sets $S\subseteq\mathbb{C}$

- *Open* if $\forall\ z_0\in U$ there is a disc around z_0 wholly in U. For example, $D_\epsilon(z), D'_\epsilon(z)$.
- *Closed* if its complement is open, and iff $S=\overline{S}$. For example $\overline{D}_\epsilon(z)$
- A *neighbourhood* of z_0 is an open set containing it.
- z_0 is a *limit point* of S if $\forall\epsilon>0, D'_\epsilon(z_0)\cap S$ is nonempty
- $\overline{S}$, the *closure* of S, is the union of S with its limit points.
- *Bounded* if there exists $M>0$ s.t. $|z|\le M$, for all $z\in S$. Otherwise it is unbounded.

Definition: Complex Sequences

Complex sequences $z_n\in\mathbb{C}$ are analogous to real ones, except index from 0.

- $z=\lim_{n\to\infty}z_n$ if for all $\epsilon>0$, there exists $N\in\mathbb{N}$ s.t. $|z_n-z|<\epsilon$ for $n\ge N$.
- z_n can be split into the combination of two real sequences a_n+ib_n, and $z=\lim_{n\to\infty}z_n$ iff $\Re(z)=\lim_{n\to\infty}a_n$ and $\Im(z)=\lim_{n\to\infty}b_n$. z_n converges iff a_n and b_n converge.
- Cauchy, Bolzano-Weierstrass, boundedness are all akin to $\mathbb{R}$
- $z\in\overline{S}$ iff there exists a sequence $z_n\in S$ s.t. $z=\lim_{n\to\infty}z_n$

Definition: Complex Functions

A complex function $f(z)$ can be written as $f(x+iy)=u(x,y)+iv(x,y)$, where $u,v:\mathbb{R}^2\to\mathbb{R}$

- For $z_0\in\overline{S}$, $a_0=\lim_{z\to z_0}f(z)$ if $\forall\ \epsilon>0$, $\exists\delta>0$ s.t. $|f(z)-a_0|<\epsilon$ when $0<|z-z_0|<\delta$.
- The limit exists iff the limits of the $\Re/\Im$ exist and equal the $\Re/\Im$ of the limit
- For $z_0\in\overline{S}, f:S\to\mathbb{C}$, $\lim_{z\to z_0}f(z)=a_0$, $w_n\in S\setminus\{z_0\}$ s.t $w_n\to z_0$, $\lim_{n\to\infty}f(w_n)=a_0$.
- Limit algebra is analogous to the reals

Definition: Continuity

- $f(z)$ is cont. at z_0 if $\lim_{z\to z_0}f(z)$ exists and equals $f(z_0)$, and cont. on a set if it is cont. at all points in that set. It is cont. at z_0 iff u,v are cont. at (x_0,y_0).
- Continuity is preserved through $+/\times/\div$(where denominator $\neq 0$) of cont. functions.
- f is cont. iff the preimage $f^{-1}(U)$ is open $\forall$ open $U\subseteq\mathbb{C}$
- Continuous functions map closed and bounded sets to closed and bounded sets

Definition: Differentiability and Holomorphicity

- $f(z)$ differentiable at z_0 if $\lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$ exists, and cont. if it is diffable.
- Diffability/holo.city preserved by $+/\cdot/\div$(where denom $\neq 0$) of diffable/holo. funcs.
- $f(z)$ is holo. at z_0 if it is diffable on a neighbourhood of z_0
- Composition preserves diffability, $(f\circ g)'(z_0)=f'(g(z_0))g'(z_0)$

Theorem: The Cauchy-Riemann Equations

If $f(z)$ is diffable at z_0 then $\frac{\partial u}{\partial x}(x_0,y_0)=\frac{\partial v}{\partial y}(x_0,y_0)$ and $\frac{\partial u}{\partial y}(x_0,y_0)=-\frac{\partial v}{\partial x}(x_0,y_0)$

- If u,v are continuously diffable (derivative is continuous) on a neighbourhood of (x_0,y_0), and satisfy the C-R equations at (x_0,y_0), then f is diffable at z_0.
- Abbrev. by defining $\partial=\left(\frac{\partial}{\partial x}-i\frac{\partial}{\partial y}\right)$, $\overline{\partial}=\left(\frac{\partial}{\partial x}+i\frac{\partial}{\partial y}\right)$. f satisfies C-R iff $\overline{\partial}f=0$.

Definition: Harmonic Functions

$h:\mathbb{R}^2\to\mathbb{R}$ is *harmonic* if it satisfies Laplace's equation, that $\forall(x,y)\in\mathbb{R}^2$, $\frac{\partial^2 h}{\partial x^2}+\frac{\partial^2 h}{\partial y^2}=0$

- Holomorphicity implies harmonicity
- For a harmonic u, v is a *harmonic conjugate of* u if $f=u+iv$ is holomorphic

Definition: Polynomials and Rational Functions

- For polynomial $P(z)=\Sigma_{n=0}^N a_n z^n$, with $a_0,\dots,a_N\in\mathbb{R}$, $P(z_0)=0$ iff $P(\overline{z_0})=0$
- For polynoms P,Q, $R:\{z\in\mathbb{C}:Q(z)\neq 0\}\to\mathbb{C}$, $R(z):=\frac{P(z)}{Q(z)}$ is a *rational function*.
- The domain of a rational function is always an open set
- Rational functions are holomorphic on $\{z\in\mathbb{C}:Q(z)\neq 0\}$

Definition: exp(z)and Complex Trig

- $\exp(z):=e^x(\cos(y)+i\sin(y))$
- $\exp(z)$ holo. on $\mathbb{C}$, $\exp(z)=\exp'(z)$
- $\exp(z+2\pi i)=\exp(z)$, so not bijective
- $\sin(z)=\frac{\exp(iz)-\exp(-iz)}{2i}$
- $\cos(z)=\frac{\exp(iz)+\exp(-iz)}{2}$
- $\sin'(z)=\cos(z)$ and $\cos'(z)=-\sin(z)$
- Periodicity, $\frac{\pi}{2}$ offset akin to $\mathbb{R}$
- $\cos^2(z)+\sin^2(z)=1$
- $\tan(z):=\frac{\sin(z)}{\cos(z)}$, $\cot(z):=\frac{1}{\tan(z)}$
- $\sec(z):=\frac{1}{\cos(z)}$, $\csc(z)=\frac{1}{\sin(z)}$
- $\sinh(z):=\frac{\exp(z)-\exp(-z)}{2}$
- $\cosh(z):=\frac{\exp(z)+\exp(-z)}{2}$
- $\sinh(iz)=i\sin(z)$, $\cosh(iz)=\cos(z)$

$\sin(x+iy)=\sin(x)\cosh(y)+i\cos(x)\sinh(y)$ and $\cos(x+iy)=\cos(x)\cosh(y)-i\sin(x)\sinh(y)$ $\sin(z+w)=\sin(z)\cos(w)+\cos(z)\sin(w)$ and $\cos(z+w)=\cos(z)\cos(w)-\sin(z)\sin(w)$

Definition: The Complex Logarithm

$\log(z):=\{w\in\mathbb{C}:\exp(w)=z\}$, $\mathrm{Log}(z):=\{\ln|z|+i\phi:\phi\in\arg(z)\}=\ln|z|+i\mathrm{Arg}(z)$

- $\log(re^{i\theta})=\ln|z|+i\arg(z)=\{\ln r+i\theta+2k\pi i:k\in\mathbb{Z}\}$
- $\log(zw)=\log(z)+\log(w)$, $\log(\frac{1}{z})=-\log(z)$
- $\mathrm{Log}_\phi(z):=\ln|z|+i\mathrm{Arg}_\phi(z)$ is holomorphic on D_ϕ and $\mathrm{Log}'_\phi(z)=\frac{1}{z}$ for all $z\in D_\phi$

Definition: Branches and Cuts

Motivated by the discontinuity line of $\mathrm{Arg}(z)$, and thus the non-holomorphicity of $\mathrm{Log}(z)$

- A branch cut is an $L\subseteq\mathbb{C}$ which is removed so that a holo. branch of a multivalued func may be defined on $\mathbb{C}\setminus L$
- $L_{z_0,\phi}:=\{z\in\mathbb{C}:z=z_0+re^{i\phi}$ for $r>0\}$. $L_{0,\phi}=L_\phi$.
- $D_{z_0,\phi}=\mathbb{C}\setminus L_{z_0,\phi}$. $D_{0,\phi}=D_\phi$. $D_{-\pi}=D$.

Definition: Complex Powers

$z^\alpha:=\{\exp(\alpha w):w\in\log(z)\}=\{\exp(\alpha\mathrm{Log}(z))\exp(i\alpha 2k\pi):k\in\mathbb{Z}\}$

- Let $\alpha,z\in\mathbb{C},z\neq 0$. If $\alpha\in\mathbb{Z}$ there is exactly one value of z^α. If $\alpha=\frac{p}{q}$, $p,q\in\mathbb{Z},p,q$ coprime, $q\neq 0$, there are exactly q values of z^α. If α is irrational or non-real there are infinitely many values of z^α.
- Let $q\in\mathbb{Z}^+$. The q values $1^{\frac{1}{q}}=1,w,\dots,w^{q-1}$, $w:=\exp(\frac{2\pi i}{q})$ are the q *roots of unity*
- Each branch of $\log(z)$ gives a branch of z^α. The *principal branch* is $z^\alpha=\exp(\alpha\mathrm{Log}(z))$
- For $z\neq 0$, $z^\alpha z^\beta=z^{\alpha+\beta}$ where the principal branch is chosen for each power.
- A branch of z^α is holo. on the D_ϕ on which Log_ϕ is holo. and $\forall z\in D_\phi$, $\frac{d}{dz}z^\alpha=\alpha z^{\alpha-1}$.

Conformal Maps and Möbius Transforms

Definition: Conformal Map

For open $U,f:U\to\mathbb{C}$, f is *conformal* if the angle between straight lines is the same as the angle between their images under f. This extends to diffable curves by considering tangents.

- A holomorphic $f(z)$ preserves angles for all $z_0\in U$ where $f'(z_0)\neq 0$

Definition: The Extended Complex Plane

The *extended complex plane* is $\tilde{\mathbb{C}}=\mathbb{C}\cup\{\infty\}$.

Let $a,b\in\mathbb{C},b\neq 0$, then $a+\infty=\infty$, $b\cdot\infty=\infty$, $\frac{b}{0}=\infty$, $\frac{b}{\infty}=0$.

Then $f(-\frac{d}{c})=\infty$ and $f(\infty)=\frac{a}{c}$. We do not define $0\cdot\infty$ or $\infty+\infty$.

Definition: Möbius Transforms

A *Möbius transformation* is a function of the form $f(z)=\frac{az+b}{cz+d}, a,b,c,d\in\mathbb{C}, ad\neq bc$

- $\frac{\lambda az+\lambda b}{\lambda cz+\lambda d}=\frac{az+b}{cz+d}$, so can always divide by $sqrt(ad-bc)$ and impose $ad-bc=1$
- The MT $f_M(z)=\frac{az+b}{cz+d}$ corresponds to $M=\left(\begin{pmatrix}cca & b\\ c & d\end{pmatrix}\right)$, with $\det(M)=ad-bc=1$.

Translation	*Rotation*	*Dilation*	*Inversion*
$f(z)=z+b$	$f(z)=az, \lvert a\rvert=1, a=e^{i\theta}$	$f(z)=rz, r>0$	$f(z)=z$
$\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}$	$\begin{pmatrix}e^{i\theta/2} & 0\\ 0 & e^{-i\theta/2}\end{pmatrix}$	$\begin{pmatrix}\sqrt{r} & 0\\ 0 & 1/\sqrt{r}\end{pmatrix}$	$\begin{pmatrix}0 & i\\ i & 0\end{pmatrix}$

Theorem: Möbius Transform Theorems

- Every MT is a composition of a finite number of translations, rotations, and dilations, and iff f does not fix the point at infinity, one inversion.
- MTs map circlines to circlines
- $f_{M_1M_2}=f_{M_1}\circ f_{M_2}$ and $f_{M^{-1}}=f_M^{-1}$
- If $f(z)$ an MT and $\exists$ distinct $z_2,z_3,z_4\in\tilde{\mathbb{C}}$ s.t. $f(z_2)=z_2, f(z_3)=z_3, f(z_4)=z_4$, $f=id$.
- For distinct $z_2,z_3,z_4\in\tilde{\mathbb{C}}$, $\exists$ a unique MT $f(z)$ s.t. $f(z_2)=1, f(z_3)=0, f(z_4)=\infty$
- For triplets of distinct points $(z_2,z_3,z_4),(w_2,w_3,w_4)\in\tilde{\mathbb{C}}$, $\exists$ a unique MT $f(z)$ s.t. $f(z_2)=w_2, f(z_3)=w_3, f(z_4)=w_4$
- MTs preserve the cross-ratio - $[f(z_1),f(z_2),f(z_3),f(z_4)]=[z_1,z_2,z_3,z_4]$

Definition: The Riemann Sphere and Stereographic Projection

Consider $\{X,Y,Z\}$ describing $\mathbb{R}^3$. Identify $\mathbb{C}$ with the plane $Z=0$ and $X+iY$ with $\{X,Y,0\}$. The *Riemann Sphere* is $S^2:=\{(X,Y,X)\in\mathbb{R}^3:X^2+Y^2+Z^2=1\}$. $N:=\{0,0,1\}$, 'north pole'. Define $\phi:\tilde{\mathbb{C}}\to S^2$ by stipulating that $\phi(z),z,N$ are collinear, and $\phi(\infty)=N$. The *stereographic projection* is $\psi:S^2\to\mathbb{C},\psi(z):=\phi^{-1}(z)$. ψ maps circles to circlines.

- $\phi(z)=(\frac{2x}{|z|^2+1},\frac{2y}{|z|^2+1},\frac{|z|^2-1}{|z|^2+1})$. $\psi(X,Y,Z)=\frac{X+iY}{1-Z}$, $\psi(N)=\infty$.

Definition: The Cross-Ratio

For distinct $z_1,z_2,z_3,z_4\in\tilde{\mathbb{C}}$, their *cross-ratio*, $[z_1,z_2,z_3,z_4]$, is the image of z_1 under the unique MT which sends (z_2,z_3,z_4) to $(1,0,\infty)$. $[z_1,z_2,z_3,z_4]=\frac{z_1-z_3}{z_1-z_4}\frac{z_2-z_4}{z_2-z_3}$.

$[\infty,z_2,z_3,z_4]=\frac{z_2-z_4}{z_2-z_3}$, $[z_1,\infty,z_3,z_4]=\frac{z_1-z_3}{z_1-z_4}$,

$[z_1,z_2,\infty,z_4]=\frac{z_2-z_4}{z_1-z_4}$, $[z_1,z_2,z_3,\infty]=\frac{z_1-z_3}{z_2-z_3}$.

Complex Integration

Definition: Complex Integration Of A Real Variable

For $[a,b]\subseteq\mathbb{R}$, $f:[a,b]\to\mathbb{C}$ is *integrable* if $u,v:[a,b]\to\mathbb{R}$ are real integrable. Then $\int_a^b f(t)\ \mathrm{d}t=\int_a^b u(t)\ \mathrm{d}t+i\int_a^b v(t)\ \mathrm{d}t$. Continuous functions are integrable. This integration is linear, respects FTC, and $\left|\int_a^b f(t)\ \mathrm{d}t\right|\le\int_b^a|f(t)|\ \mathrm{d}t$

Definition: Contours

For $z_0,z_1\in\mathbb{C}$, a *(parametrised) curve* Γ connecting them is a continuous func $\gamma:[t_0,t_1]\to\mathbb{C}$ for $t_0,t_1\in\mathbb{R},t_0<t_1$, s.t. $\gamma(t_0)=z_0$ and $\gamma(t_1)=z_1$.

- γ can be split into $\Re/\Im$ parts how you think it can be.
- Γ is *regular* if γ is continuously diffable and $\gamma'(t)\neq 0$ for all $t\in(t_0,t_1)$
- Γ is a closed and bounded subset of $\mathbb{C}$
- For regular Γ, the *arclength* $\ell(\Gamma):=\int_{t_0}^{t_1}\left|\gamma'(t)\right|\ \mathrm{d}t=\int_{t_0}^{t_1}\sqrt{x'(t)^2+y'(t)^2}\ \mathrm{d}t$
- For Γ an arc of a circle of radius r through an angle θ, $\ell(\Gamma)=r\theta$
- A curve Γ is a *contour* if it is a finite union of regular curves where the endpoints join up.
- A contour is *simple* if it has no self-intersections, except possibly at the endpoints.
- A *loop* is a simple closed contour
- A loop is *positively-oriented* if the interior is on the left as we move along the loop.

Definition: Contour Integrals

Given $z_0,z_1\in\mathbb{C},\Gamma$ a regular curve connecting them, and $f:\Gamma\to\mathbb{C}$ continuous, we define the *integral of f along Γ* by $\int_\Gamma f(z)\ \mathrm{d}z=\int_{t_0}^{t_1}f(\gamma(t))\gamma'(t)\ \mathrm{d}t$

- Contour integration is linear (can pull out factors, split sums)
- For two parametrisations of Γ, $\gamma:[t_0,t_1]\to\mathbb{C}$ and $\tilde{\gamma}:[\tilde{t_0},\tilde{t_1}]\to\mathbb{C}$, if there exists injective diffable $\lambda:[\tilde{t_0},\tilde{t_1}]\to[t_0,t_1]$ with $\lambda'(t)>0\forall t\infty[\tilde{t_0},\tilde{t_1}]$ s.t. $\tilde{\gamma}(t)=\gamma(\lambda(t))\ \forall\ t\in[t_0,t_1]$, then the integrals along γ and $\tilde{\gamma}$ are equal.
- If Γ is parametrised by $\gamma:[0,1]\to\mathbb{C}$, then integrating $-\Gamma$ parametrised by $\tilde{\gamma}:[0,1]\to\mathbb{C},\tilde{\gamma}(t)=\gamma(1-t)$ gives the negative of the integral along Γ.
- A contour is *closed* if $\gamma(t_0)=\gamma(t_1)$
- A *contour integral* is the sum of the integrals along the component regular curves.

Definition: Domains

$D\subseteq\mathbb{C}$ is a *domain* if D is open and every two points in D can be connected by a contour lying wholly in D.

- If $u:D\to\mathbb{R}$ is diffable, with $\frac{\partial u}{\partial x}=\frac{\partial u}{\partial y}=0$ on D, then u is constant on D.
- A continuous f on D has an *antiderivative* F on D if $F'(z)=f(z)$ for all $z\in D$.
- The integral of a cont. $f:D\to\mathbb{C}$, with an antideriv on D, along a closed Γ in D, is 0.
- D is *simply-connected* if $\mathrm{Int}(\Gamma)\subseteq D$ for all loops $\Gamma\subseteq D$.

Lemma: Path-Independence

For a domain $D\subseteq C$ and continuous $f:D\to\mathbb{C}$, the following are equivalent:

i f has an antiderivative F on D
ii $\int_\Gamma f(z)\ \mathrm{d}z=0$ for all closed Γ in D
iii All contour integrals are independent of the path Γ, and depend only on the endpoints

Theorem: Jordan Curve Theorem

Let Γ be a loop in $\mathbb{C}$. Then Γ defines two regions in $\mathbb{C}$ with Γ as their common boundary: a bounded domain, the *interior* of Γ, $\mathrm{Int}(\Gamma)$; and an unbounded domain, the *exterior* of Γ, $\mathrm{Ext}(\Gamma)$.

- $\mathbb{C}=\mathrm{Int}(\Gamma)\cup\Gamma\cup\mathrm{Ext}(\Gamma)$

Theorem: Cauchy Integral Theorem

Let Γ be a loop and f be holomorphic inside and on Γ. Then $\int_\Gamma f(z)\ \mathrm{d}z=0$.

- If f is holo. on a simply-connected domain D, then f has an antiderivative on D
- If $z_0\in\mathbb{C}$ and Γ is a loop s.t. $z_0\notin\Gamma$, then $\int_\Gamma\frac{1}{z-z_0}\ \mathrm{d}z=2\pi i$ if $z\in\mathrm{Int}(\Gamma)$ and 0 otherwise

Theorem: The Deformation Theorem

For loops $\Gamma_1,\Gamma_2\subseteq\mathbb{C}$, and f holomorphic on $\Gamma_1,\Gamma_2,\{\mathrm{Int}(\Gamma_1)\setminus\mathrm{Int}(\Gamma_2)\}\cup\{\mathrm{Int}(\Gamma_2)\setminus\mathrm{Int}(\Gamma_1)\}$, $\int_{\Gamma_1}f(z)\ \mathrm{d}z=\int_{\Gamma_2}f(z)\ \mathrm{d}z$

Theorem: Cauchy Integral Formula

For a loop Γ, $z_0\in\mathrm{Int}(\Gamma)$, and f holo. inside and on Γ, $f(z_0)=\frac{1}{2\pi i}\int_\Gamma\frac{f(z)}{z-z_0}\ \mathrm{d}z$

Theorem: Generalised Cauchy Integral Formula

Given a loop Γ, $z_0\in\mathrm{Int}(\Gamma)$, and f holo. inside and on Γ: f is infinitely differentiable at z_0 and for $n\in\mathbb{N}$, $f^{(n)}(z_0)=\frac{n!}{2\pi i}\int_\Gamma\frac{f(z)}{(z-z_0)^{n+1}}\ \mathrm{d}z$

- Can be rearranged to calculate contour integrals if appropriate derivatives are known
- For $z_0\in$ domain D, $R>0$ s.t. $\overline{D}_R(z_0)\subseteq D$, f holo. on D, and $M>0$ an upper bound for $|f(z)|$, then for all $n\in\mathbb{N}$, $\left|f^{(n)}(z_0)\right|\le\frac{n!M}{R^n}$

Theorem: Morera's Theorem

If $D\subseteq\mathbb{C}$ a domain, $f:D\to\mathbb{C}$ cont., $\int_\Gamma f(z)\ \mathrm{d}z=0$ for all loops Γ inside D, then f is holo.

Theorem: Liouville's Theorem

If f is holo. on $\mathbb{C}$ and bounded (ie $\exists M>0$ s.t. $|f(z)|\le M\forall z\in\mathbb{C}$), then f is constant.

- For $D\subseteq\mathbb{C}$ a domain, $z_0\in D$, $R>0$ s.t. $\overline{D}_R(z_0)\subseteq D$, and f holo. on D; $f(z_0)=\frac{1}{2\pi}\int_0^{2\pi}f(z_0+Rexp(it))\ \mathrm{d}t$
- Given the above and f s.t. $\max_{z\in\overline{D}_R(z_0)}|f(z)|=|f(z_0)|$, $f(z)$ is constant on $\overline{D}_R(z_0)$.

Theorem: The Maximum Modulus Principle

Let $D\subseteq\mathbb{C}$ be a domain, and f be holo. and bounded on D. Then if $|f(z)|$ achieves its maximum on D, f is constant on D.

- If a func is holo. on a bounded domain, and cont. up to and including the boundary, it attains the maximum of its modulus on the boundary.

Theorem: Maximum/Minimum Principle For Harmonic Functi

Let $D\subseteq\mathbb{R}^2$ be a domain and $\phi:D\to\mathbb{R}$ be harmonic s.t. ϕ is bounded above or below on D by $M>0$, and $\phi(z_0)=M$ for some $z_0\in D$. Then ϕ is constant on D.